REMOTE SENSING AND ITS APPLICATIONS IN INDIA

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1.0 Introduction

Exploration of Earth's natural resources is vital for the socio-economic development of any country. The state-of-the-art technology of space based earth observation systems offers timely and accurate information on various natural resources such as land, water, forests, mineral resources etc. Though the potentials of orbiting space platforms were realized at the time of launching of the very first satellite, the Russian "Sputnik" in 1957, the remote sensing application especially for earth resources came into its own from 1972 with the launch of "Earth Resources Technology Satellite" later renamed as Landsat by National Aeronautics and Space Administration (NASA) of United States of America (USA). The various application studies carried out in India and elsewhere in the world proves beyond doubt that remote sensing is a powerful tool for mapping/inventorying, monitoring and managing of natural resources due to the inherent advantages of synoptic viewing, repetitive imaging, capability to study inaccessible areas, relatively low cost and near real time availability of data. This paper explains in brief about remote sensing and its applications in India.

2.0 Principles of Remote Sensing

Remote sensing is mainly concerned with the measurement or acquisition of information about an object without being in physical contact with the object under study. The term remote sensing is used to methods, which use electromagnetic energy as the means for collection of information about the object. Depending on its physical features and properties, the earth surface reflects or reradiates or emits different kinds and amounts of electromagnetic energy in various wavelengths. The measurement of reflected or radiated or emitted electromagnetic radiation forms the basis for understanding the characteristics of Earth's surface features (Colwell, 1983 and Sabins, 1997). In passive remote sensing system, the sensors operating in different selected spectral bands onboard the airborne/space borne platforms measure the naturally radiated or reflected energy from the earth's surface features. An active remote sensing system supplies its own source of energy to illuminate the objects, and measures the reflected/ back-scattered energy returned to the system.

Only selected portions of the electromagnetic spectrum those can pass through the Earth's atmosphere with relatively little attenuation are used for remote sensing purposes (Fig.1). Each object of the earth's surface features reflects, radiates or emits different kinds and amounts of electromagnetic energy in different regions of the electromagnetic spectrum. These typical responses are used to distinguish the

objects from one another. Typical spectral response curves for selected features are given in Fig.2. These spectral reflectance curves represent the percent of incident light reflected by materials as a function of wavelengths.

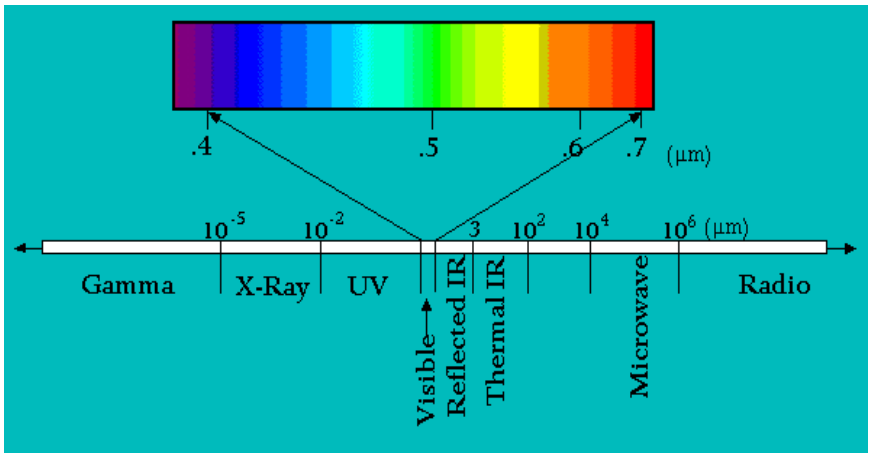


Figure-1: Electromagnetic Spectrum

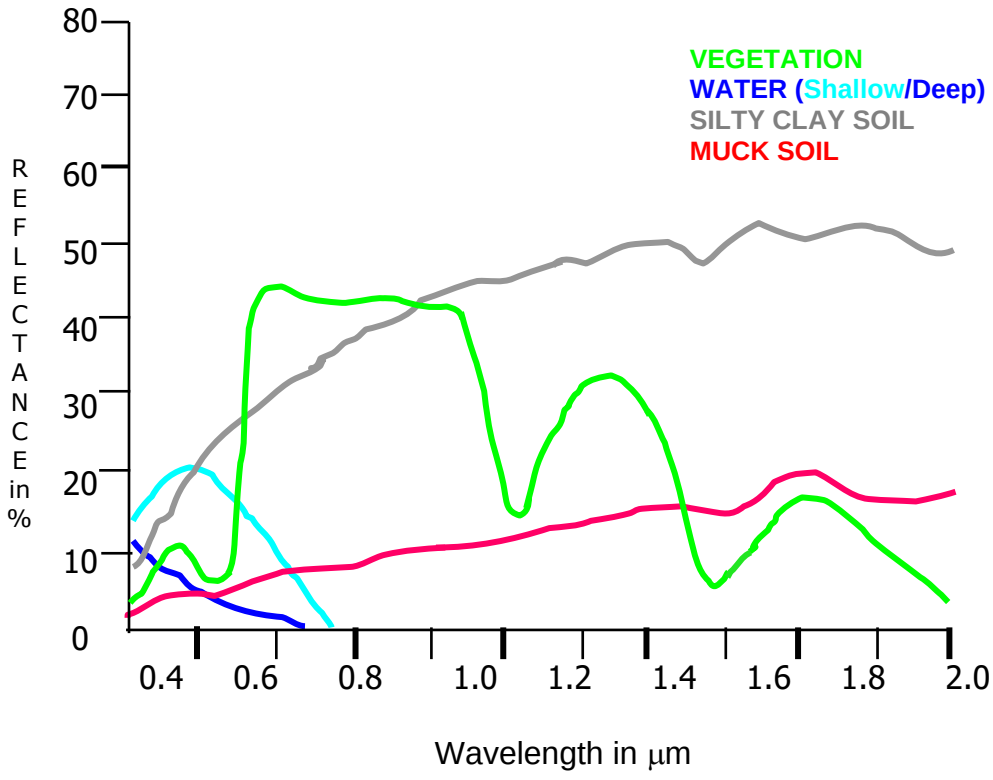


Figure-2: Typical Spectral Reflectance Curves

3. 0 Satellites and Sensors

Earth Resources Technology Satellite (ERTS) later renamed as Landsat -1 was the first remote sensing satellite to be launched by NASA for surveying, mapping and monitoring of earth resources. Realizing the potentials of this emerging technology many other countries like France (satellite series named SPOT), India (satellite series named IRS) have entered into this venture. So far six satellites in the series of Landsat have been launched of which first three are of first generation satellites and carried Return Beam Vidicon (RBV) and Multispectral Scanner (MSS) imaging sensors and the second generation satellites carried apart from MSS advanced imaging sensor called Thematic Mapper (TM) with 30m spatial resolution and seven spectral bands (including Short Wave Infra Red and Thermal bands). The latest Landsat 7 has a sensor called Enhanced Thematic Mapper (ETM) with spatial resolution of 15m. Three satellites in the series of SPOT launched by France provided data in MSS and panchromatic spectral bands (20 and 10 m spatial resolution) with normal and/or stereoscopic mode. Indian Remote Sensing Satellite - IRS-1A and IRS-1B, the indigenously developed satellites have been put into orbit in 1988 and 1991 respectively and both of them provided good quality data continuously (with spatial resolution of 72 m & 36 m for- Linear Imaging Self Scanning Sensors: LISS -I and LISS –II respectively).

European Remote Sensing satellites (ERS-1 & 2) of European Space Agency (ESA) and RADARSAT of Canada provides Radar data (which can collect data in all weather conditions and in day & night). These data are found to be useful in studying crops, snow, geology etc. The Radar Interferometry data generated using ERS-1&2 are found to be useful in earthquake studies.

Unlike geostationary satellites, these remote sensing satellites are sun synchronous (same local equatorial crossing time of the descending node enabling the study of natural resources at various seasons under the same illumination condition), and polar orbiting type with a repetitivity cycle of 16 to 26 days enabling repeated collection of data of the same place at the same local time for continuous monitoring of the earth's resources. The imaging payloads of these satellites operate in different spectral bands (parts of electromagnetic spectrum), spatial resolution (the smallest area on the ground being sensed by the sensors), and radiometric resolution (number of grey/reflectance levels which is distinguishable).

Remote sensing surveys carried out till date use data from remote sensing satellites such as Landsat, IRS & SPOT etc. Sometimes information is supplemented by aerial photos/airborne sensors. The resolutions offered by these satellites broadly fall in four categories - low, medium, high and very high. Low resolution (70-80 m pixel size) is offered by Landsat MSS and IRS LISS I sensors. Medium resolution (20-40 m) is offered by Landsat - TM, IRS-LISS II, LISS III and SPOT HRV MLA sensors, high resolution (5-10m) is available from SPOT HRV PLA, IRS-1C/1D PAN and Resoucesat LISS-IV sensors. IKONOS & Quick Bird satellites offer very high resolution (4m&1m and 2.4m & 0.61m).

During the last 3-4 years tremendous progress has been achieved in acquiring data with very high spatial resolution, hyper spectral data, stereo data, increased revisit capability etc. This started with the launch of commercial satellite missions like IKONOS of Space Imaging Corporation of USA in 2000. IKONOS with 1m spatial resolution in panchromatic and 4m in multi spectral bands revolutionized the remote sensing applications. With the launch of Quick Bird in 2001, data with 61cm (in panchromatic and 2.4 m in multispectral mode) are made available. Orbview -3 launched in 2003 also provides 1m & 4m data. All these satellites provide stereo images and high radiometric resolutions. Now it is possible to go up to mapping on 1:2000/4000 scales. Height information/contours can also be derived using stereo pairs of these data and it is found that 2-4 m contour can be derived. Hyperspectral data from sensors like MODIS (Moderate Resolution Imaging Spectroradiometer) on Terra and Aura satellites and ASTER (Advanced Space borne Thermal Emission and Reflection Radiometer) on Terra satellites are found to be useful in studying various features.

Airborne Laser Terrain Mapper (ALTM) the latest addition in the remote sensing technology using laser beams can provide resolution up to 15cm and is found to very useful for taking up detailed studies of canal alignment, contour extraction etc.

3.1 Indian Remote Sensing Satellites (IRS)

India's first operational indigenously developed Remote Sensing Satellite IRS-1A in the IRS series was successfully launched on March 17, 1988. IRS-1A was placed in the sun-synchronous orbit of 904 km with equatorial crossing time of the descending node being at 10.25 AM, which enables the study of natural resources at various seasons under the same illumination condition. The repetitivity cycle of IRS-1A was 22 days.

The major payloads of IRS-1A consisted of two types of imaging sensors, which operated in push broom scanning mode using Linear Imaging Self Scanning Sensor (LISS). In this mode of operation each line of the image is electronically scanned by a linear array of detectors called Charged Coupled Devices (CCD) consisting 2048 elements. The first type of imaging sensor provided a spatial resolution of 72 meters and designated as LISS-I and the other type consisted of two separate imaging sensors providing spatial resolution of 36 meters and designated as LISS-IIA and LISS-IIB. LISS-I provided a swath of 148 km on ground while LISS-IIA and LISS-IIB provided a composite swath of 145 km on ground.

The spectral bands selected for the IRS-1A are almost similar to that of first four bands of Landsat TM sensor. As follow-on to IRS-1A, the second operational Remote Sensing Satellite, IRS-1B in the IRS series was launched successfully on August 29, 1991. IRS-1B is functionally identical to that of IRS-1A and phased in such a way as to provide a combined repetitivity cycle of 11 days, as against 22 days provided by each of them. The IRS-1A and IRS-1B data products are being used extensively in various application projects like geological mapping, ground

water prospect zone mapping, drought management, water resources management, land use/land cover mapping, etc.

IRS- 1C/1D launched in 1995/1997 have sensors such as (i) LISS-III (with spatial resolution of 23 m and SWIR- Short Wave Infrared band) (ii) PAN (Panchromatic band with 5.8 m spatial resolution) and (iii) WiFS (Wide Field Sensor- with 188m spatial resolution). The data from these sensors are being extensively utilized for various application projects. IRS-P6 (RESOURCESAT) launched in October 2003 has sensors like LISS-IV (5.8 m spatial resolution in multi-spectral mode) AWiFS (Advanced Wide Field Sensor - with spatial resolution of 56 m) and LISS-III (identical to one in IRS-1C/1D). The future satellite missions include Cartosat 1/2 with PAN sensor having spatial resolution 2.5m/1m, Radar satellite Risat etc.

4.0 Image Interpretation and Analysis

The satellite data products are available to the resource scientists either in the form of photographic products such as films, dispositive, paper prints of various scales or digital data stored in the form of floppies and CD-ROM or through Internet. The desired information can be extracted from these data products through visual interpretation and/or digital image processing techniques. In practice, both visual interpretation and digital image processing techniques are complimentary to each other. For large areas and spectrally homogeneous scenes digital image processing method may provide a quick and cost effective means of image analysis. On contrary for smaller areas and spectrally heterogeneous scenes visual interpretation method may be more suitable because of the need for basic image characteristics, visual perception etc.

4.1 Visual Interpretation Techniques

Visual interpretation of remote sensing images, for extracting desired information could be achieved in an efficient and effective way by using several basic interpretation keys or elements (Sabins, 1997). The basic interpretation keys are (i) tone (ii) texture (iii) pattern (iv) shape (v) size and (vi) location or association. All these interpretation elements are qualitative attributes and they are subjective depending on the experience and personal bias of an interpreter. The characteristics of each of the interpretation keys are given below:

- **Tone/Colour** refers to relative brightness of objects on the black and white image or refers to the combination of hue, saturation and intensity in colour images. The tone in the black and white images is expressed as different shades or levels of grey. For example water, vegetation, soil etc. have distinct tonal variations. Without tonal differences, the shape, pattern and texture could not be discerned. Tonal variations depend upon the reflectance properties of surface materials, terrain slope, illumination condition, etc.

- **Texture** is the frequency of change and arrangement of tones/colour on an image. It is produced by a spatial arrangement/aggregation of features that may

be too small to discern individually on the image. The visual impression of roughness or smoothness created by some objects is a valuable clue in image interpretation. The texture is normally referred as fine, medium or coarse and also as stippled or mottled.

- **Pattern** on an image result from regular repetition of tonal variations and textures. The repetition of certain general forms or relationships is characteristic of many natural and man made objects. Arrangement of vegetation, topographic features, drainage networks, different rock types are typical examples of pattern. The most commonly used patterns are (i) the drainage networks, which have orderly association with geologic structure and (ii) rock types and outcrop patterns, which provide clues to geologic structures.

- **Shape** refers to the general form, configuration or outline of individual objects. The shape of some object is so distinctive that their image may be identified solely from this criterion. Stadiums, race course, golf course, tanks, geological features such as folds, igneous intrusions, faults, deltas, volcanic cones, sand dunes etc. are having specific shape that helps the interpreter for delineation.

- **Size** of an object is one of the most useful clues to its identity and should be discussed in the context of the image scale. For example streams versus rivers can be differentiated based on the size as the criterion in their identification.

- **Location** or **association** of objects in relation to other features gives important clues on the identity of an object. A circular feature in an area of metamorphic rocks is unlikely to be a salt plug, irrigated crop near a tank; sand beaches along the coastline are the examples of association.

Among the above elements tone and texture are the functions of brightness, contrast and resolution of the image while size, shape and pattern are dependent on the scale of an image.

4.2 Digital Image Processing Techniques

Remotely sensed data corresponds to the different earth features collected by the sensors and stored in cartridge/floppy/CD-ROM in the form of rasters or tiny equal areas or picture elements or pixels and are arranged in regular lines and columns. The pixels represent brightness value having a specific Digital Number (DN) value, which depends on the energy reflected by the earth surface in a specific wavelength or band or channel. Most of the sensors on-board the satellites operate in different discrete wavelengths. Therefore, each one of the earth features is sensed by the sensors simultaneously and provides a set of DN values. The DN value of each pixel is ranging from zero for black to some higher value for white, based on the radiometric resolution (eg: 6 bit data represents 0-63 grey level, 7 bit data represents 0-127 grey levels and so on). The availability of remotely sensed data in digital form helps in carrying out digital image processing with the aid of computers. The digital image processing techniques provide flexibility in data handling due to the fact

that the digital data can be numerically manipulated by using an equation or set of equations to get the desired details in the graphic display monitor or pictorial form for further interpretation.

There are a number of procedures/methods available for image data manipulation (Hord, 1982; Jensen, 1986; Lillesand and Kiefer, 1994; Schowengerdt, 1997; and Mather, 1999). However, they can be broadly grouped into three categories, viz., (i) Image rectification and restoration also called as pre-processing, (ii) Image enhancement, and (iii) Image classification.

- Image rectification and restoration/pre-processing operations are intended to eliminate or correct the distortions or errors caused due to geometric distortions, radiometric distortions, presence of noise in the data, etc. The standard products made available to the interpreters are pre-processed. Therefore, generally data can be directly used for image enhancement and classification.

- Image enhancement operations (Contrast stretching, Band Subtraction, Brightness Index, Vegetation Index, Band Ratioing, Principal Component Analysis, Decorrelation Stretching, Filtering, Intensity-Hue-Saturation (IHS), resolution merging etc) are being applied to image data to get the enhanced output for subsequent visual interpretations. The enhancement techniques provide better feature exhibitance, i.e., increases the visual distinction between features contained in a scene. The enhanced outputs can be seen in the display monitor or can be recorded in the pictorial form as B&W and/or colour composite images.

- Image classification operations are essentially meant to analyze the remotely sensed data with quantitative assessment. The classification of the remotely sensed data can be carried out either without a priori knowledge about the features present in the scene (unsupervised classification) or with a priori knowledge about the terrain features (supervised classification).

5.0 Remote Sensing Applications

Remote sensing data (satellite/aerial) are being used to map/monitor/survey/manage various natural resources. The technique is being used to study land use/cover, soil, geology, geomorphology, agriculture, forestry, water resources, urban areas, wastelands, coastal zone, glaciers, volcanoes, earthquakes, floods, drought, oceans etc. In India large number of remote sensing application projects have been carried out during the last 3 decades. A unique system called National Natural Resources Management System (NNRMS) has been set up to optimally manage country's natural resources using Remote Sensing and conventional techniques. Brief on various major applications is given below.

5.1 Agriculture

Agriculture is the one area where remote sensing has been used successfully (i) to

estimate crop acreage, (ii) to assess the crop damage (iii) to estimate production etc., Pre-harvest Crop Acreage and Production Estimation (CAPE) of major crops (wheat, rice, rapeseed/mustard, cotton, groundnut, bajra and sugarcane crop) using remote sensing data is one of the major ongoing projects being carried out in collaboration with state and central agencies. Pre-harvest acreage and production estimation of major food grains, oilseeds and fiber crops are generated regularly and provided to Department of Agriculture in advance (2 months). The project is carried out in major crop growing areas in 15 states (AP, Assam, Bihar, Gujarat, Haryana, HP, Karnataka, Orissa, Maharashtra, MP, Punjab, Rajasthan, Tamil Nadu, UP & W. Bengal).

A major project on Forecasting Agricultural output using Space, Agrometeorology and Land Observations (FASAL) has been taken up on pilot scale basis for selected states. The project envisages an integrated approach to CAPE on lines similar to MARS (Monitoring of Agricultural Statistics with Remote Sensing in the European Countries). FASAL addresses all important issues such as improved timeliness, better accuracy and extended coverage of crops, besides providing more than one assessment during a crop season and having a provision for alternate forecast methodology in case of non-availability of cloud free satellite data.

Projects have been carried out to determine the actual area under mulberry and to conduct pilot studies on detection and estimation of area under ‘tasar’ and ‘muga’ plants and to assess the suitability of districts in the selected states for taking up sericulture.

GIS database on Indian Farmers Fertilizer Cooperative (IFFCO) infrastructure, road network, settlement, natural resource layers, weather and historical data have been created. A Decision Support System (GeoFARMS) for fertilizer movement - for entire country has been developed. Horticulture crops are also being studied using remote sensing and GIS.

5.2 Water Resources

One of the first applications of remote sensing in water resources area is for surface water inventory. Presently remote sensing data are being used for number of applications viz Irrigation Water Management, Command area surveys/monitoring/management, Dam/reservoir site assessment, Canal Alignment, Snow melt Runoff Forecasting, Snow/Glacier studies, Surface water bodies mapping/monitoring, River/water body pollution assessment, Reservoir sedimentation/Siltation assessment & Capacity evaluation, Water Quality Mapping and Monitoring etc.

5.2.1 Command area studies

Command area development is an area where remote sensing can contribute significantly. To monitor the impact of Command Area Programme (CAD) implementation on major irrigation projects since 1985, Ministry of Water Resources

(MoWR) requested ISRO/DOS to carryout satellite based Inventory, monitoring and performance evaluation of Irrigation systems of 14 selected irrigation projects covered in 5 states in two phases. The irrigation project performance with respect to Rabi/Summer season crop productivity was evaluated using satellite data of three-time period scenario (e.g., 1986-87, 1990-91, 1998-99). Thirteen CAD projects namely, Upper Tapi, Purna, Jayakwadi, Krishna, Girna and Bhima in Maharastra, Chambal (Kota) in Rajasthan, DVC System, Mayurakshi and Kanshabati command in West Bengal, Sriram Sagar, Nagarjun Sagar in Andhra Pradesh, Jamuna in Assam have been studied.

A major project to assess saline and / or alkaline affected areas; waterlogged areas using multi-temporal satellite data in 333 Commands of major & medium irrigation projects has been taken up recently at the behest of CWC in Rajasthan, Haryana, Bihar, Jharkand, Karnataka and Goa States.

5.2.2 Irrigation Water Management

Management of water supplies for irrigation in command areas requires information on total demand and its distribution. The current satellite remote sensing capabilities for irrigation water management include end-of-season evaluation of canal command areas at the disaggregated level and diagnostic analysis of problem distributaries to enable follow-up corrective management.

Satellite remote sensing has been used for base line inventory of irrigated area, cropping pattern, crop condition and productivity assessment in irrigation systems, monitoring irrigation status through the season, optimum design of crop cutting experiments etc.

Simultaneous monitoring of crop performance through the irrigation season is a critical element in effective water management in the irrigation projects. Monitoring of crop condition is essential to ensure that the irrigation requirements are adequately met by canal releases. The 5 days repetitive coverage of WiFS sensor provides the necessary surveillance capability across the command area through the season. The inventory capability is significantly enhanced with the higher spatial resolution offered by LISS III sensor and the additional SWIR band has improved crop condition assessment. WiFS sensor helps in concurrent monitoring during the irrigation season, generating near real-time information on the sowing progress and crop condition for appropriate irrigation water delivery. Data from the Advanced WiFS data with 56m spatial resolution from IRS P6 will further improve this activity. The stereo coverage from PAN sensor providing contour maps of 10 m interval or better, helping in preliminary analysis of canal alignment and land/infrastructure development. The 1m & 61cm resolution data from IKONOS/Quick Bird etc; could be used for delineating details of canal network and field level crop inventory.

Many irrigation projects in India and elsewhere suffer from adverse impacts of excessive irrigation and poor drainage. Satellite remote sensing has helped in mapping areas affected by salinity/alkality and water logging. In many irrigation

projects in India, the areal extent and severity of water logging and salinisation have been mapped. The severity of such soil limitation can be either directly detected through remote sensing or seen by the impact on crop productivity. A major project to map and monitor the saline/water-logged areas in the command areas of 800 reservoirs has been launched by Regional Remote Sensing Service Centres (RRSSCs) recently at the behest of Central Water Commission (CWC).

5.2.3 Reservoir Capacity Evaluation

Storage capacity of the reservoir decreases due to periodical deposition of sediment discharge, which causes reduction in irrigation intensity in a command area. Information on sedimentation process is vital for taking up appropriate measures in controlling the sediment inflow and judicious management of the available storage in the reservoir. The reduction of storage capacity of a reservoir due to sedimentation is indicated by the reduction in contour area at a given elevation. Monitoring the changes in water spread, which is nothing but contour area at different operating levels of the reservoir will help to assess the quantum of sediment load that had settled down in the reservoir. Remote sensing satellites, which provide data on water spread at regular intervals of time, could be used successfully, for estimating the water spread and monitoring the seasonal changes in water spread. The reservoir sedimentation study in terms of storage loss estimation of some reservoirs namely; Bhadra, Malaprabha, Ghataprabha, Krishnarajasagara, Mahi Bajajsagar, Hirakud, Ukai-Kadana has been carried out using multi-date satellite data. Recently a major project has been launched to estimate the capacity loss due to sedimentation of 20 reservoirs in the country.

5.2.4 Snowmelt Runoff Forecasting

Snowmelt runoff modeling in different Himalayan basins is being routinely carried out using satellite data. National Oceanic and Atmospheric Administration/Advanced Very High Resolution Radiometer (NOAA/AVHRR) have been used in 23 large river basins for real time mapping and monitoring of snow cover. Since 1994, seasonal snowmelt runoff forecast for the Sutlej river basin is being operationally provided to Bhakra Beas Management Board by 1st week of April every year based on analysis of daily observed NOAA satellite AVHRR data from February to June period and subsequent updating. True forecast is obtained with an error of maximum ±10% with respect to actual measured inflow.

5.2.5 Snow and Glacier Investigations

Snow and glacier investigation through satellite data has been carried out in the Himalayan region. Snow cover in parts of Himalayas has been monitored using high repetitivity WiFS data at regular 5-day interval. Retreat of eight glaciers of Basapa valley has been studied using PAN stereo data. Map showing crevasses of Siachen Glacier using PAN data at 1:15,000 scale is being prepared. 1m & 61cm spatial resolution data will be very useful in detailed study of snow and glaciers. Ministry of

Environment and Department of Space have proposed a major project for studying the snow & glaciers of the Himalayas.

5.2.6 Water Quality Mapping and Monitoring

Surveillance of water quality is an important activity for multiple uses such as irrigation, water supply, etc., and also for the protection of aquatic and shoreline environment. Water bodies are polluted through sedimentation, industrial effluents, municipal sewage, agricultural run off, eutrophication and oil pollution. Remote sensing of water quality can complement ground efforts in mapping and monitoring point and non-point pollution sources, the influx and dispersal of pollutants in the aquatic environment and consequent impact such as algal blooms and weed growth. Growth of aquatic weeds and algal blooms indicating the eutrophication status of water bodies can be mapped from satellite data. Aquatic weeds are responsible for deterioration of water quality and reduction of aquatic life. A number of tanks in India infested with aquatic weeds have been mapped using satellite data. High resolution data from IKONOS, Quick Bird and hyper spectral data from MODIS are found to be useful in assessing the water quality. Oil spill monitoring (in ocean), turbidity assessment etc have also been carried out successfully using remote sensing.

5.3 Drought Assessment and Monitoring

The periodic droughts are very common in India due to failure or irregular onset of monsoon as the Indian agriculture is mostly dependent on rainfall with an estimate of around 94 million ha rain fed area out of 142 million ha arable land. Therefore, the drought combating strategies like drought prone area programmes have been taken up and are showing encouraging results. In this context ISRO/DOS has developed a national/regional system for assessment and monitoring of agricultural drought using satellite data based vegetation index to provide timely information on drought severity. Towards this, National Agricultural Drought Assessment and Monitoring System (N-ADAMS) have been launched in 1987 to characterize the severity of drought in the country. Since then, N-ADAMS Bulletins on monthly and seasonal crop conditions depicting agricultural drought condition are issued at States/District levels, based on vegetation indices and ground based information for 14 states and sub district levels for 6 States during the kharif season.

5.4 Flood Mapping and Damage Assessment

Remote sensing provides comprehensive, reliable and timely information on flood inundated and drainage congested areas, extent of damage to crops, structures etc., River configuration, silt deposits and vulnerable areas of bank erosion and flood risk zones could be studied using remote sensing data. However comprehensive flood monitoring would call for integration of ground measurements and remote sensing data and flood plain characteristics including topography. The flood risk zone maps are updated with high spatial resolution data and digital elevation models based on close contour information. Flood damage vulnerability analysis requires integration of the information on satellite derived physical damage and socio-economic data.

Since 1987, all major flood events of the country have been mapped in near real time and statistics on crop area affected and number of marooned villages generated. Near real time flood monitoring is being carried operationally in Brahmaputra (Assam), Kosi/ Ganga (Bihar), Indus (J & K), Godavari (AP) and Mahanadi (Orissa) river basins using optical and microwave data. Studies have been carried out for delineation of flood risk zone in Bramhaputra and Kosi river basin.

However, presence of cloud cover during the flood-monitoring period is a limitation for optical remote sensing data. In this case, microwave data from ERS1/2 and Radarsat satellites are effectively supporting the study in the persistent cloud cover during the flood events. The WiFS sensor with 5 days repetitivity on board in IRS 1C/D satellite, provide unique opportunity to map and monitor flood events frequently over large river basins. The detailed impact assessments in larger scale are being implemented through LISS III and PAN sensors. The availability of very high-resolution data (61cm/1m) is helping in studying the flood at cadastral level. The planned RISAT (Radar Imaging Satellite) of ISRO will provide the radar data in future.

5.5 Forestry

Forest cover mapping using satellite data is one of the first studies done in the early 80's by National Remote Sensing Agency (NRSA) Hyderabad. Subsequently Forest Survey of India (FSI) took up this activity and regularly prepares forest cover maps once in two years using satellite data on 1:250,000 scale (recently FSI has switched over to 1:50,000 scale) for monitoring the forest cover changes and planning conservation measures.

Forest fire mapping/damage assessment, wildlife habitat evaluation, forest type mapping/management plan preparation, forest encroachment studies are also carried out at different parts of the country. A major project has been recently launched by the Ministry of Environment & Forests to map forest types of the country on 1:50,000 scale.

5.6 Bio- diversity Mapping

A major project 'Biodiversity Characterisation at Landscape Level' to prepare Bio-resource Data Base for four important bio-diversity rich areas viz., NE Region, Western Himalaya, Western Ghats and Andaman & Nicobar Islands of the country through remote sensing and GIS was taken up subsequent to Biodiversity Convention held during 1992 at Rio De Janeiro and has been completed. The project was carried out in two phases. The phase-I is aimed at bio-resource mapping and prioritisation of areas for bio-diversity conservation, while under the phase-II detailed assessment of bio-diversity and their distribution pattern in 10 important bio-diversity rich sites was carried out. Important medicinal plants, which have a significant role in the preparation of drugs to cure cancer, asthma and other life threatening diseases, have been mapped in the high altitude Himalayas. A

Biodiversity Information System (BIS), containing exhaustive database on species, has also been developed which is web enabled. Similar work has been taken up in Central India, Eastern Ghats and Mangroves of east coast.

5.7 Environmental Impact Assessment

Satellite remote sensing provides the synoptic overview to generate objective base line information for proposed projects and in evaluating impact in the case of existing projects. Analysis of time series of satellite data enables monitoring changes before, during and after implementation projects like dams, reservoirs etc. Inventory of land cover/land use in the reservoir submergence area, particularly the forest and agricultural cropped areas has been carried out using satellite data. Irrigability, mapping of command area, through integration of satellite derived and ground measured data, would help in appropriate allocation and application of water to minimize water logging and salinization. In India, satellite data is used potentially in environmental monitoring of Idukki hydel project, Kerala, Western Ghat areas including Nilgiri Biosphere Reserve, large-scale river valley projects of Narmada and the Tehri dam in Himalayas. The satellite remote sensing survey in these regions clearly brought the degradation of vegetation cover due to biotic interference as well as long spells of adverse climatic condition. High-resolution data is of tremendous help in assessing the impact before during and after the dam/reservoir/canal construction etc.

Environmental Impact Assessment of Industries/Mining/Thermal Power Plants is also carried out in the country. Suitability of sites for setting up of chemical industries (like distilleries) was assessed in Karnataka, UP & Punjab. Studies have been carried out to assess the ecological feasibility of proposed mining activity and the impact of ongoing mining (Bailadila & Keonjhar Iron Ore Mines; Baphlimali, Surguja & Chintapalli Bauxite Mines). Thermal Power Plants like Farakka, WB; Ramagundam AP; Korba & Rhiand, MP were ecologically monitored. The change in forest cover due to mining in Sandur area, Karnataka has also been assessed.

5.8 Wasteland Mapping

13 categories of wastelands (degraded/under used/un cultivable land) like gullied or ravinous land, upland with or without scrub, waterlogged/marshy areas, degraded forests etc were mapped in the country in 5 Phases, during 1986–2000, on 1:50,000 scale. Digital database of all the maps have been created. Total wastelands in the country are estimated to be 64 Mha, of which 46 Mha is culturable and 18 Mha is un-culturable. Wasteland maps have been recently updated using satellite data of 2003.

5.9 Ocean/Coastal Studies

Under the programme known as Marine Satellite Information Services (MARSIS),

several activities for developing capabilities for retrieval of oceanic parameters and monitoring of coastal zone, and providing such data to down stream users have been undertaken. Potential Fishing Zone (PFZ) advisories based on integration of Sea Surface Temperature (SST) derived from NOVAA-AVHRR data and ocean colour from Ocean Colour Monitor (OCM) on IRS P4 were generated thrice in a week (Tuesday, Thursday and Saturday) for the entire Indian coast and is disseminated to the fishermen. Presently Indian National Centre for Ocean Information Services (INCOIS) is providing the PFZ forecast through Bulletin Boards, Fax, Telex etc. Plans are afoot to disseminate through digital audio broadcasting.

Mapping of coastal zone features such as land use/land cover, wetland/land form, mangroves and shoreline changes on 1:250,000 and 1:50,000 scale covering the entire coast of India has been completed. Methodologies for retrieval of various ocean parameters such as surface winds, wave forecast model, internal waves and wave spectrum have been developed.

Coral reef maps on 1:50,000 scale have been prepared for Gulf of Kachchh, Gulf of Mannar, Palk's Bay, Lakshadweep and Andaman & Nicobar islands. Mapping of features between High Tide Line (HTL) and Low Tide Line (LTL) and land use features for a strip of 500 m from HTL, on 1:25,000 scale, under Coastal Regulation Zone (CRZ) was also carried out. Ministry of Environment & Forests has recently launched a project to study coastal areas, corals and mangroves.

5.10 Watershed Planning and Management

Judicious management and conservation of soil and water resources on watershed basis is a pre requisite for sustaining the productivity. Characterization and prioritisation of watersheds are essential steps towards the integrated management. Watershed characterization involves measurement of parameters of geological, hydrogeological, geomorphological, hydrological, soil, land cover /land use etc. Remote sensing via aerial and space borne sensors can be effectively used for watershed characterization and assessing the watershed priority, evaluating problems, potentials and management requirements and periodic monitoring. Remote sensing data greatly facilitates mapping of land use/cover, geology, soils and other features of watershed, which would assist in the study of land use pattern, water potential, degradation etc. This along with ground based information can be used for assessing the land capability classes, irrigation suitability classes, potential land uses, suitable water harvesting measures, monitoring the effects of watershed conservation/development measures, correlation of run off and sediment yields from different watersheds and monitoring land use changes and land degradation.

Remote sensing based Integrated Mission for Sustainable Development (IMSD) project, based on integration of various themes and collateral information, carried out in about 84 million ha in 175 districts of the country has been successful in improving the ground water level, increasing the cropping intensity along with the net returns from the fields and finally conserving the land and water resources. Under National Watershed Development Programme for Rain fed Areas (NWDPRA) 20

watershed in Karnataka and 110 watersheds throughout the country were monitored to assess the impact of various developmental programmes, which had been undertaken in those watersheds.

Remote sensing techniques are also providing significant information inputs towards calibration, validation and use of various hydrologic models like Soil Conservation Service (SCS) model for rainfall run-off prediction, Sediment Yield Index (SYI) and Universal Soil Loss Equation for prioritization of watersheds, etc.

5.11 Desertification Monitoring

The studies have taken up jointly with Ministry of Agriculture for monitoring desert in the Thar desert. Using satellite data, desertification indicators such as sand dune movement, change in green bio-mass, soil erosion, wind erosion, water logging, salinity & alkality, frequency of dust storms and precipitation pattern are monitored to assess the impact of control measures being undertaken in Thar desert. An integrated approach, involving the assessment of each of these indicators and their impact on natural resources has been taken up for understanding the process of the desertification in the region.

A Convention on Combating Desertification (UNCCD) was organized by the United Nations in 1994. One of the important initiatives taken during the convention was to develop a framework for the formulation of Regional Action Plan (RAP) and substantiating plans of action by means of establishing Thematic Programme Networks (TPNs). Six thematic programme areas have been identified under RAP. The first one, Thematic Programme Network (TPN -1) is on Desertification Monitoring and Assessment. Asian countries participating in TPN -1 include China, India, Iran, Japan, Jordan, Kazakhstan, Kyrgystan, Kuwait, Lebanon, Mongolia, Oman, Pakistan, Philippines, Syria, Tajikistan, Turkmenistan, Uzbekistan and Yemen. China has been identified as the host country to coordinate TPN -1 activities. In India, Space Applications Centre has been identified as the national focal institution to coordinate TPN-1 activities within the country and establish the national network for desertification monitoring and assessment. Ministry of Environment and Forests, Govt. of India coordinates UNCCD activities in the country.

Towards generating information for TPN-1 a project titled ‘Desertification Monitoring and Assessment using Satellite Remote Sensing and GIS: A Pilot Study’ has been taken up by SAC at the behest of MOEF to evolve and standardize methodology.

Under the pilot study, mapping is being carried out on 1:50,000 scale adopting a three level hierarchical system of classification based on (i) Land use/Land Cover (ii) Processes of Degradation and (iii) Severity of Degradation. The UNCCD definition of desertification (as given in article 1) – “ land degradation in arid, semi arid and dry-sub humid areas resulting from factors including climatic variations and human activities” has been followed in the pilot study and accordingly areas/sites in arid, semi arid and dry-sub humid regions of the country viz Pali & Dausa Districts of

Rajasthan; Ratlam Dist. MP; Bellary District, Karnataka; Balia Dist. UP; Panchkula Dist. Haryana; Mehboob Nagar Dist. (one watershed) AP; Theni Watershed, TN; Kathua, Padam, Hanle, Leh, Panamic, Diskit (in J&K); Chamba, Keylong and Lahul Spiti (in HP) watersheds have been covered under the study.

Several organizations/departments/universities like Central Arid Zone Research Institute (CAZRI), National Bureau of Soil Survey and Land use Planning (NBSS&LUP), All India Soil & Land Use Survey (AIS&LUS), National Thematic Mapping Organization (NATMO), Institute of Remote Sensing, Anna University, Remote Sensing Application Centres of MP/Haryana/J&K, Universities of Rajasthan/Jammu/JNU and ISRO/DOS centers are involved in the work with SAC as the lead centre.

5.12 Geology and Mineral Resources

One of the first to use remote sensing are geologists. The remote sensing data was used to study structural geological features like faults/folds/fractures etc, geomorphology, lithology and in turn used in mineral/oil exploration, ground water exploration & development, engineering geology, environmental geology neotectonics, seismotectonics etc., Under Project Vasundhara major part of South India was studied and detailed data base was created. Geological Information System for the states of Maharashtra & Chhattisgarh has been developed. Landslide hazard zonation maps on 1:25,000 scale for the pilgrim/tourist routes of Uttaranchal and Himachal Pradesh have been prepared.

Seismotectonic studies of Gujarat, Maharashtra, Karnataka, Tamil Nadu and other places also have been carried out. Eruptions of Barren Island Volcano in 1991 & 95 have been monitored. Atomic minerals, diamond and oil prospective areas have been identified using remote sensing. A number of several other remote sensing base studies have also been carried out in the country for geological applications.

In Jharia Coal Field underground coal fire prone/risk zones were delineated using airborne and satellite thermal data. Thermal anomalies as deep as 70 metres were detected and suggested suitable remedial measures

5.13 Ground Water Prospect Zone Mapping

Remote sensing data helps in delineating the potential ground water occurrence zone with less time and cost effective manner than the conventional methods. The thematic layers on geology, geomorphology, drainage, soil and land use generated from satellite data, are essential in identifying the ground water prospect zones. Remote sensing technology has been used for preparation of hydrogeomorphological maps of the entire country on 1:250,000 scale using satellite images under the National Drinking Water Mission, launched by Government of India in 1986 as an effort to provide sources of safe drinking water to 1.6 lakh problematic villages. Under this mission, broad groundwater prospective zones have

been delineated for further follow-up through hydro geological and geophysical methods to provide potable drinking water to the problematic villages.

Subsequently, under Rajiv Gandhi National Drinking Water Mission (RGNDWM), launched in 1999, preparation of ground water prospects map is being carried out on 1:50,000 scale using high resolution satellite data (IRS LISS-III data). Andhra Pradesh, Karnataka, Kerala, Madhya Pradesh, Chattishgarh and Rajasthan states are covered in phase -I and Jharkhand, Himachal Pradesh, Orissa and Gujarat states are being covered under phase-II. The main objective of this project is to enable scientific source finding of drinking water for all the non-covered (NC) and partially covered (PC) habitations (around 2 lakh) and to identify suitable recharge structures. Around 2000 maps have been prepared so far and have been submitted to Department of Drinking Water Supply of Ministry of Rural Development and the State Ground Water and other related departments.

So far (till June 2004) around 65,000 wells have been drilled with about 90% success in Andhra Pradesh, Karnataka, Kerala, Madhya Pradesh, Chattishgarh and Rajasthan states. Around 3500 recharge structures also have been built in these states.

5.14 Urban

Using remote sensing data, detailed developmental plans for selected cities like Mumbai, Bangalore, Delhi, Hyderabad, Ahmedabad in the country have been carried out. City guide maps on 1:10,000 scale have also been prepared. Base map updation, urban land use change and growth monitoring, master plan development, facilities management, taxation applications, urban information systems are the other major studies being planned. Under National Urban Information System (NUIS) 136 towns of the country will be studied. New capitals areas for Chattishgarh and Jharkhand were identified using remote sensing.

5.15 Infrastructure

Remote sensing data along with GIS & GPS have been extensively used to find out suitable corridors/alignment of roads, railways, pipelines, canals, tunnels etc. One of the first such studies is the alignment of ring road for Bangalore City carried out in 1990's. The Mysore – Bangalore infrastructure corridor, gas pipeline between Mangalore – Bangalore, gas pipeline grid of Gujarat, Oil pipelines in Maharashtra, Gujarat and West Bengal etc; are some of the projects where in Remote Sensing inputs were used. Selection of suitable sites for bridges (both road/railway) in UP and Bihar, assessment of suitability for airports, harbours and major industries are the other projects carried out. With the availability of very high resolution data from Quick Bird, IKONOS and other satellites it is possible to go up to 1:4000/2000 scale which will help in a big way for all the infrastructure projects.

5.16 National (Natural) Resources Information System (NRIS)

NRIS forms the core information system for the NNRMS and is oriented to aid decision-makers at national, regional, state and district levels. The NRIS is a set of spatial, GIS-based database of thematic and non-spatial data organised into interlinked/networked hierarchy so as to cater to free flow of resources information. Implemented for 30 Districts and 4 State nodes. Contains 22 spatial & 8 non-spatial layers of information – pertaining to natural resources as well as socio-economics – with necessary analysis modules. NRIS for Orissa state is ready, for UP, Punjab, Maharashtra, Gujarat, Karnataka, MP, Haryana, Mizoram work is in progress. For Delhi, Chandigarh, A&N Islands work is in progress also progressing well for the other 22 districts. Specific user oriented query shells has been developed.

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